Literature Review: Multi-Agent Rainwater Management for Urban Flood Mitigation

1. Introduction

This review asks what existing research offers for a team of classical-AI agents that must reroute and store rainwater during a storm when sensors are noisy and drainage routes close unexpectedly. The project is a simulated multi-agent system for a synthetic Jeddah-like district (a graph of ten zones: main roads, residential, industrial, school and coastal-park areas, and two storage basins, each with a fixed capacity and drainage direction). Three agents coordinate through one shared environment model: Rainfall Monitoring (Bayesian updating of noisy sensor readings), Storage Management (rule-based thresholds that reroute inflow from a nearly full basin to the basin with the most free capacity) and Route/Drainage (A* routes around blocked connections, still in development). No machine learning is used. Success is measured against a fixed-drainage baseline by number of flooded zones, runoff stored and replanning success after injected faults.

The review covers ten peer-reviewed conference and journal papers drawn from the IEEE Xplore exports supplied with the project; nine are from 2021 onward. They are grouped by approach: flood-management analysis and decision support [1]–[5], search-based path planning [6]–[8], dynamic hazard-aware routing [9], and blackboard-style multi-agent coordination [10]. Most exported records were off-topic (consensus control, power grids, network routing), so the set is small and uneven, and that unevenness is itself part of the finding.

2. Existing AI Approaches and Methodologies

Theme A: flood analysis and decision support. The flood-specific papers analyse systems rather than control them. Piro et al. survey real-time control of stormwater drainage, predictive models, decision support and early-warning systems [1]. Kamashki and Hdeib run hydrological and hydraulic assessments of the Al Lawzi drainage system in Bahrain under extreme-rainfall scenarios [2], and Sonavane et al. simulate Pune's drainage in EPA SWMM, finding surcharge at junctions J-3, J-9, J-11 and J-21 and recommending larger drains [3]. Hdeib ranks nature-based solutions for arid cities with multi-criteria decision analysis (MCDA) and finds rainwater harvesting best, ahead of infiltration trenches and permeable pavements [4]. Ramachandran et al. use a random forest on rainfall, growth rate and soil type to site storm drains [5]; it is machine learning, so it is included only as a contrast. All five treat the storm as a fixed input and produce a static design or ranking.

Theme B: search-based path planning. Ali and Das implement an A* variant on a 15×15 grid, chosen because A* guarantees an optimal route, and add a velocity-obstacle method for moving obstacles [6]. Yu et al. combine an improved Theta* with B-spline smoothing for a vehicle, tested in two simulated scenarios with static and moving obstacles [7]. Scheffe et al. recast receding-horizon control as graph search over a finite-state automaton, prove recursive feasibility and run two vehicles in real time [8]. Together they show that grid search and horizon-limited planning are real-time capable, which supports the route agent (horizon-based storage planning as in [8] is a possible later extension, not part of the current design), but every obstacle is a physical object and not a hazard with a cost.

Theme C: dynamic hazard-aware routing. Priority-based Discrete-time Dynamic Routing (PDDR) models multi-source, multi-destination evacuation as a network-flow problem that prioritises high-risk locations and replans as the hazard changes, and it beats a no-priority variant (DDR) and static greedy routing (SGR) [9]. This is the closest analogue to the route agent, but it addresses building evacuation, not rainwater.

Theme D: multi-agent coordination. Yang and Liu coordinate satellite agents through a blackboard with serial or parallel worker activation. Their mean fitness is up to 20% higher than the best centralised algorithm and negotiation is more than 80% cheaper than contract-net [10]. It supports coordination through shared state, as in the project's shared environment model, although the problem solver is a hybrid metaheuristic and not classical search.

3. Critical Comparison

The studies split on one axis: those closest to the flood problem use no classical AI, and those that use search or planning never see rainfall. Table I summarises this. Figures come from the abstracts only; where an abstract gives none, the cell says so.

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
[1]	Real-time control, predictive models, decision support (survey)	Urban stormwater drainage and early warning	Qualitative review; no test cases	No implementation or metrics
[2]	Hydrological/hydraulic modelling	Al Lawzi lake system, Bahrain; extreme-rainfall scenarios	Capacity weaknesses identified; no figures in abstract	Static design analysis; no runtime adaptation
[3]	EPA SWMM simulation with GIS	Pune drainage; past critical storms	Surcharge at J-3, J-9, J-11, J-21; larger drains advised	Pre-2021; fixed events; no agents
[4]	Multi-criteria decision analysis	Nature-based options for arid cities	Rainwater harvesting ranked first; score not in abstract	Weights and data; static ranking
[5]	Random forest (ML, out of scope)	Storm-drain siting by district	Parameter influence analysed; no accuracy in abstract	ML; static siting; official data scarce
[6]	A* grid search; velocity-obstacle avoidance	15×15 grid, moving obstacles	Reported efficient and reliable; no numbers in abstract	Small grid; no hazard costs; no uncertainty
[7]	Improved Theta* with B-spline	Vehicle; two static/moving-obstacle scenarios	Effective in simulation; no numbers in abstract	Kinematics-specific; two scenarios; no

Ref	AI Technique	Problem / Test Cases	Evaluation & Result	Limitation
				baseline
[8]	Graph search over finite-state automaton; receding horizon	Two-vehicle trajectory planning, lab and simulation	Real-time; recursive feasibility proved	Only two agents; no uncertainty
[9]	Network-flow dynamic routing with priorities	Multi-source evacuation in closed spaces	PDDR beats DDR and SGR; margins not in abstract	Buildings, not floods; hazard assumed known
[10]	Blackboard multi-agent planning; metaheuristic	Distributed satellite mission planning	Fitness up to 20% above centralised; negotiation down more than 80%	Satellites; offline; not classical search

Table I. Summary of reviewed studies.

Three points emerge from the comparison. First, evaluation is weakest where relevance is highest: the flood papers [1]–[5] compare nothing against a baseline, while only [9] and [10] benchmark against alternatives, and [6]–[8] validate in simulation alone. Second, uncertainty is largely missing. None of the abstracts describes noisy observations handled by probabilistic reasoning, and [9] assumes the hazard state is known. Third, adaptation is treated as an obstacle problem, not a resource problem: the planners [6]–[8] avoid obstacles but none allocates a limited resource such as reservoir capacity, which is where the project's storage agent and its overflow limit matter.

4. Limitations and Research Gap

The unmet requirement is a single loop: none of the ten papers closes the path from noisy sensing, through hazard-aware routing, to constrained storage within one storm event. It breaks into five gaps:

1. **Static versus in-event decisions.** Flood studies [1]–[5] size or rank infrastructure for a design storm; none replans while the event unfolds.
2. **No probabilistic sensing in the loop.** Uncertainty is absent from the planning papers [6]–[9]. Bayesian updating of noisy rainfall readings feeding a planner is not covered by any paper in this set.
3. **Obstacles, not hazards.** A* and Theta* [6], [7] treat blocked cells as binary; none weights routes by flood risk or reopens them when water recedes.
4. **Routing without storage.** Planning horizons appear in [8], but none allocates reservoir capacity over a horizon, and the arid-region study [4] ranks storage options without simulating them.
5. **Weak evaluation.** Few studies state a baseline, run counts or fault injection, so claims such as "effective" or "outperforms" cannot be compared.

The project addresses some of these gaps, not all of them:

Gap	Status in project	Basis
1. Static vs in-event	Addressed in simulation	Replanning during a simulated storm; not field-tested
2. No probabilistic sensing in the loop	Addressed by design	Bayesian updating feeds the other agents
3. Obstacles, not hazards	Open	A* treats closed routes as binary; no flood-risk weights
4. Routing without storage	Partial	Threshold rules only; horizon-based allocation deferred
5. Weak evaluation	Addressed by design	Fixed-drainage baseline, 30 runs per scenario at 10, 30 and 60 mm/h, with 20% sensor dropout, Gaussian noise (σ = 5 mm/h) and one to three route closures

5. References

1. P. Piro, M. M. Saleh, B. Pirouz, M. Turco, and S. A. Palermo, "Smart and innovative systems for urban flooding risk management," in *Proc. 2023 Int. Conf. Inf. Commun. Technol. Disaster Manage. (ICT-DM)*, 2023, pp. 1–4, doi: 10.1109/ICT-DM58371.2023.10286951.
2. R. Kamashki and R. Hdeib, "Towards resilient stormwater management system. Application of AI-Lawzi Lake stormwater system," in *Proc. 2024 ASU Int. Conf. Emerg. Technol. Sustain. Intell. Syst. (ICETSIS)*, 2024, pp. 182–187, doi: 10.1109/ICETSIS61505.2024.10459470.
3. N. Sonavane, V. A. Rangari, M. L. Waikar, and M. Patil, "Urban storm-water modeling using EPA SWMM — a case study of Pune city," in *Proc. 2020 IEEE Bangalore Humanitarian Technol. Conf. (B-HTC)*, 2020, pp. 1–5, doi: 10.1109/B-HTC50970.2020.9297900.
4. R. Hdeib, "Nature-based solutions for sustainable urban drainage: Implications for arid region urban resilience," in *Proc. 2025 Int. Conf. Decis. Aid Sci. Appl. (DASA)*, 2025, pp. 2337–2343, doi: 10.1109/DASA68193.2025.11499033.
5. R. Ramachandran, A. D, S. D, A. B. Anil, A. Babu, and H. MD, "Harnessing QGIS and random forest analysis for precision storm drain reinforcement," in *Proc. 2024 Int. Conf. Emerg. Res. Comput. Sci. (ICERCS)*, 2024, pp. 1–5, doi: 10.1109/ICERCS63125.2024.10895146
6. M. Ali and S. Das, "Mobile robots with dynamic obstacle avoidance," in *Proc. 2023 IEEE 3rd Int. Conf. Sustain. Energy Future Elect. Transp. (SEFET)*, 2023, pp. 1–6, doi: 10.1109/SeFeT57834.2023.10245465.
7. H. Yu, X. Wang, and W. Sun, "An improved Theta*-based trajectory planner for autonomous vehicle with obstacle avoidance," in *Proc. 2021 4th IEEE Int. Conf. Ind. Cyber-Phys. Syst. (ICPS)*, 2021, pp. 613–618, doi: 10.1109/ICPS49255.2021.9468211.
8. P. Scheffe, M. V. A. Pedrosa, K. Flaßkamp, and B. Alrifaee, "Receding horizon control using graph search for multi-agent trajectory planning," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 3, pp. 1092–1105, May 2023, doi: 10.1109/TCST.2022.3214718.

9. S. N. P, J. Bapat, and D. Das, "Priority based discrete time dynamic routing algorithm for emergency evacuation," in *Proc. 2021 IEEE Int. Conf. Electron., Comput. Commun. Technol. (CONECCT)*, 2021, pp. 1–6, doi: 10.1109/CONECCT52877.2021.9622540.
10. Y. Yang and D. Liu, "Distributed imaging satellite mission planning based on multi-agent," *IEEE Access*, vol. 11, pp. 65530–65545, 2023, doi: 10.1109/ACCESS.2023.3289964.

Appendix: Verification Log

Ref	Title	Venue	Year	DOI	Full text opened
[1]	Smart and Innovative Systems for Urban Flooding Risk Management	ICT-DM	2023	10.1109/ICT-DM58371.2023.10286951	Yes
[2]	Towards Resilient Stormwater Management System	ICETSIS	2024	10.1109/ICETSIS61505.2024.10459470	Yes
[3]	Urban storm-water modeling using EPA SWMM	B-HTC	2020	10.1109/B-HTC50970.2020.9297900	Yes
[4]	Nature-Based Solutions for Sustainable Urban Drainage	DASA	2025	10.1109/DASA68193.2025.11499033	Yes
[5]	Harnessing QGIS and Random Forest Analysis	ICERCS	2024	10.1109/ICERCS63125.2024.10895146	Yes
[6]	Mobile Robots with Dynamic Obstacle Avoidance	SEFET	2023	10.1109/SeFeT57834.2023.10245465	Yes
[7]	An Improved Theta*-based Trajectory Planner	ICPS	2021	10.1109/ICPS49255.2021.9468211	Yes
[8]	Receding Horizon Control Using Graph Search	IEEE Trans. Control Syst. Technol.	2023	10.1109/TCST.2022.3214718	Yes
[9]	Priority based Discrete time Dynamic Routing Algorithm	CONECCT	2021	10.1109/CONECCT52877.2021.9622540	Yes
[10]	Distributed Imaging Satellite Mission Planning Based on Multi-Agent	IEEE Access	2023	10.1109/ACCESS.2023.3289964	Yes